

Raghav Verma

Backend / Platform Software Engineer | Distributed Infrastructure, Service Mesh & Recommendation Systems
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SUMMARY

Software engineer with 5+ years at Salesforce building distributed infrastructure, backend services, and ML platform systems. Delivered zero-downtime migrations across 100+ functional domains, a recommendations engine serving 2.8B+ peak-load personalizations, and \$1M+ annualized GMV impact across 50+ enterprise storefronts.

WORK EXPERIENCE

Salesforce Jan 2021 – Present
Software Engineer | Java, Python, Kubernetes, Istio, OpenSearch, PyTorch | New York, NY

- MCIS Personalization, Recommendations & ML Platform** Aug 2024 – Present
- Owned OpenSearch-based vector retrieval and contextual bandit reranking for the recommendations engine across 50+ enterprise storefronts, serving 2.8B+ personalizations at peak Black Friday load with p99 latency under 350ms.
 - Drove \$1M+ annualized GMV impact by redesigning catalog exclusion logic, anchor-item matching, and per-dataset operational controls, increasing CTR by 22% while reducing irrelevant recommendations by 7%.
 - Reworked evaluation data loading from single-file reads to parallel chunk processing, reducing per-thread memory usage from ~2GB to under 400MB.
 - Architected an offline NDCG@K evaluation framework across thousands of user profiles and 15+ production regions, enabling model tuning that improved Goal-Based Recommender NDCG by up to 3x.

- Ingress Gateway & Service Mesh Platform** Jan 2021 – Aug 2024
- Led zero-downtime migration of Salesforce's two largest ingress services onto a standardized release platform across dev, test, stage, and production, rolling out to 100+ functional domains with no customer-facing availability gaps.
 - Built a developer-facing service analyzer that automated validation of service configs (ports, DNS, target port usage), reducing on-call tickets by over 20% across backend services.
 - Shipped certificate lifecycle features (enrollment states, 5xx error handling) for vault-backed auth systems, improving reliability across customer-facing flows.
 - Mentored 3 junior engineers, served as Scrum master for a 9-person team, and managed on-call operations for Kubernetes/Istio infrastructure across AWS-managed clusters.

- Salesforce | Software Engineer Intern (Database Systems)** May 2020 – Aug 2020
- Improved database IO performance by 20% by building an index drop/restore system for team-managed database environments.

- Amazon, Seattle, WA | Software Engineer Intern (AWS, DynamoDB, Python)** May 2019 – Aug 2019
- Designed and shipped an end-to-end vendor management application on AWS (DynamoDB, Lambda), from requirements through production deployment; adopted by the team and still in use post-internship.

PROJECTS

- BetEdge | Full-Stack Sports-Market Analytics Platform | GitHub**
FastAPI, Postgres, SQLAlchemy 2.0, React 19/TypeScript, Docker
- Built live odds ingestion and vig-removed probability modeling to estimate fair market probabilities from US sportsbook lines.
 - Designed an expected-value ranking engine and bankroll ledger for auditable simulated and manual wager tracking.
 - Developed a backtest harness scoring strategies on forecast quality (Brier score, log loss, calibration) and portfolio metrics (ROI, max drawdown).

EDUCATION

University of California, Berkeley | Bachelor's in Computer Science & Data Science May 2020

TECHNICAL SKILLS

Languages	Java, Python, Go, SQL
Backend / Distributed Systems	OpenSearch, gRPC, service mesh, DynamoDB, MongoDB, REST, vector search
Infrastructure	Kubernetes, Istio, Envoy, AWS, Docker, Linux
CI/CD & Build	Jenkins, Spinnaker, Maven, Git
ML / Data	PyTorch, Spark, Dask, contextual bandits, recommendation systems, Jupyter
Observability	Grafana, Splunk, internal monitoring platforms